

Machine-Learning Application for Fast 3-Phase Black Oil Reservoir Simulation

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Abstract: Reservoir simulation is essential for field development and well planning through the estimation of pressure, saturation distribution, and production rate decline prediction. One of the primary objectives of the simulation is to predict the hydrocarbon recovery from a reservoir in a given period of time under study. When the field-scale reservoir model is highly heterogeneous and complex, then the commercial reservoir simulators employ a large number of grid blocks to model the reservoir, and this necessitates the solving of a large number of finite-difference discretized mass balance equations. Solving these equations requires a large amount of computational time and detailed information about the reservoir properties like porosity and permeability. Also, when the fluid properties are highly sensitive to pressure, then a large number of iterations are performed at each time step to predict the fluid properties, which as a result, makes the traditional reservoir simulation process computationally expensive. In this article, a novel methodology is proposed in which a machine learning (ML) model designed using a random forest is used to reduce the number of iterations performed at each time step, which expedites the simulation process. This is done by predicting reservoir fluid properties like formation volume factor, density, and viscosity using ML at the first iteration, which provides a close initial estimate of the parameters needed to solve the finite-difference equations. This first step reduces the number of iterations required to reach convergence and solve for pressure and saturation without changing the conventional finite-difference equations. The applicability of the proposed methodology has been verified on a 3-phase black oil reservoir model. The study's novelty is in the development of a novel ML-based 3-phase black oil reservoir simulator that can speed up the reservoir simulation process while conserving the standard fluid flow physics involved, making the model computationally more efficient. DOI: [10.1061/JLEED9.EYENG-6060](https://doi.org/10.1061/JLEED9.EYENG-6060). © 2026 American Society of Civil Engineers.

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Introduction

Reservoir Simulation

Reservoir simulation is one of the most efficient methods for modeling fluid flow in a reservoir through the integration of data from a variety of fields, such as geology, petrophysics, geophysics, and reservoir and production engineering. The resulting model is historically matched utilizing pressure and production measurements in order to improve the geological understanding and prediction capabilities. The matching must be performed properly in order to predict the production of a field with reliable accuracy since each hydrocarbon reservoir has distinct geological properties, reservoir properties, and drive mechanisms (Mohaghegh 2014). The matching involves the creation of an adequate reservoir model and determining the production behavior of the reservoir by adjusting the geological parameters (initial and boundary conditions) and control parameters, then predicting the variation of the time-dependent

parameters such as production rate decline, pressure, and saturation distributions (Illarionov et al. 2022). Reservoir simulation is performed in the following stages: (1) the primary objective is determined; (2) all the reservoir data are gathered and validated; (3) a reservoir model is built; (4) the reservoir model is history matched; and (v) the reservoir model is employed for performing predictions of the reservoir's production behavior. The traditional reservoir simulation involves a number of complicated steps, which not only require a large amount of computational time to execute but also make the process complex. Therefore, to reduce the complexities involved, researchers worldwide have been trying to make the simulation process faster and computationally more efficient (Ertekin et al. 2001). In many of the previous works cited in the literature survey section, solving the discretized mass balance equations was a major challenge since it necessitates the involvement of high-performance equipment and excessive computational time, making the process computationally less efficient. From the literature survey, it is clear that the excessive computational time, high-performance equipment, and algorithms not considering the fluid flow physics were major drawbacks of the previously published works.

Literature Survey

Reservoir properties data were employed as a training data set in a machine learning (ML) model to develop a three-dimensional meta-model of a development unit. The proposed methodology requires executing a large number of simulations, and in order to execute them smoothly, high-performance computational equipment is necessary to make the process computationally more

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efficient (Simonov et al. 2018). Numerical simulation of the naturally cracked carbonate was performed using Smart Proxy. The process did not result in a significant reduction in simulation time, and its applicability was limited to a few field case models as it involved the oversimplification of the fluid flow physics (He et al. 2016). In another study (Zhang et al. 2016), the fast marching method (FMM) and calculated diffusive time of flight (DTOF) were employed to provide an estimation of the production behavior of the shale gas reservoir. The proposed methodology is significantly faster than the traditional reservoir simulation and uses a modified physics-based reservoir simulation algorithm, which restricts its application only to unconventional reservoirs (Zhang et al. 2016). Xue et al. (2019) simulated the hydraulic fractures occurring with the natural fractures through the combination of FMM, local grid refinements (LGRs), and an embedded discrete fracture model (EDFM). The methodology is much faster than the traditional reservoir simulation but is applicable only to unconventional reservoirs (Xue et al. 2019).

The pressure and saturation distribution across the reservoir at each time step were predicted by Mohaghegh et al. (2012) with reasonable accuracy by applying a grid-based surrogate reservoir model (GSRM). The study proved to be computationally less efficient due to the need for high-performance computing equipment (Mohaghegh et al. 2012). Cornelio et al. (2021) trained an artificial neural network (ANN) model using the formation parameters, completion parameters, and production responses of the Bakken shale to transfer learning and predict the production for the Eagle Ford shale by using its input parameters (Cornelio et al. 2021). Teng et al. (2018) predicted the gas production in shale gas reservoirs using FMM with the succession of steady state (SSS). The proposed method proved to be faster and more stable than the traditional implicit finite-difference-based reservoir simulation due to the requirement of a much smaller number of linear equations to be solved at each time step (Teng et al. 2018). Production history data were gathered by Monfaredi et al. (2020) to build a fast reservoir simulator, which was used for history matching and to predict the reservoir's production behavior in a shorter period. The method proved to be computationally less efficient as it involved the execution of many complicated steps that required high-performance computing resources (Monfaredi et al. 2020).

Mohmad et al. (2020) performed history matching by integrating data preparation, quality assessment, spatiotemporal database formulation, reservoir model design, ANN, and data-driven analytics. However, the method was observed to have limited application due to the lack of involvement of fluid flow physics. Kam et al. (2017) used the water-cut, gas-oil ratio, and bottom-hole pressure data to perform streamline-based 3-phase history matching. The method was significantly faster than the traditional reservoir simulation but was applicable to only a few field-scale models. Park et al. (2020) trained an ML model using physics-based reservoir simulation input and output data to facilitate the development of an unconventional field. The proposed methodology was faster than the traditional reservoir simulation and gave prediction results with reasonable accuracy.

Some researchers have employed ML modeling to predict the production rate decline and EUR estimated ultimate recovery (EUR) of multiple wells. Bhattacharyya and Vyas (2022a, b, c) predicted the production rate decline and EUR of several Eagle Ford shale oil wells and Bakken shale oil wells using an ML-based model to determine the decline curve parameters of the stretched exponential decline model (SEDM). Although these articles used similar ML-based methodology, the data sets belong to completely different oil fields, confirming the applicability of the proposed methodology to multiple oil fields. LaFollette et al. (2013) conducted

multivariate analysis of production well data collected from the Bakken Light Tight Oil Play. They observed that the longer lateral wells gave lesser production per foot of lateral length. In a later study, LaFollette et al. (2014) collected production well data from the Eagle Ford Formation in South Texas for multivariate analysis, and the trends in reservoir fluid parameters were reported. The study showed that gas-oil ratio (GOR) is one of the most important variables. It was observed that while production rates increase with the completed lateral length, the production per unit completed length decreases as the completed length increases. It was also found that increasing the amount of proppant used for job completion increases the productivity in terms of maximum monthly production.

Ansari and Govindrajana (2024) investigated the use of hot water injection for improving the production performance in heterogeneous reservoirs, as an emerging energy demand due to population and industry growth. The study utilizes American Petroleum Institute (API) gravity and reservoir rhythm to investigate the impact of hot water flooding. To determine the degree of hot water flooding, a careful investigation of the permeability fluctuation is also conducted. To assess hot water flooding situations, an enhanced, fully coupled energy and multiphase flow model was developed to simulate the evolution of fluid and rock properties in two-dimensional reservoirs. Mwakipunda et al. (2023) examined the impact of water flooding-based infill drilling optimization in the Clair Oil Field to optimize the net present value (NPV) using 3D numerical models. Finding the best well sites and operational characteristics (production restrictions) to maximize cumulative oil output was a key objective of this reservoir management study. The aspects that can be taken into account in infill drilling to increase oil reserves were illustrated by theoretical concepts and mathematical models with field data performance. Zheng et al. (2023) developed a 3D thermal-hydraulic-mechanical (THM) model to simulate heat extraction from a granite reservoir, examining the impact of two interacting crossed-rough cracks on the heat recovery capacity. Thermal power and reservoir recovery rate were used as indicators to assess the heat extraction performance, and the effects of fracture surface roughness and the angle between the two rough fractures are investigated. The simulation data showed that when the fracture aperture varies, so does the geometry of the fluid flow line on the rough fracture surface. Xie et al. (2022) replicated reservoir deformation and fluid flow behaviors simultaneously using a discrete fracture model (DFM). Numerical solutions for the strain and pore pressure were obtained using the finite-element method (FEM). A perfect agreement was achieved when the numerical simulator was validated using commercial software. The findings show that the evolution of the formation's physical properties is significantly impacted by the interaction of the flow and geomechanics. In the early stages, secondary fractures (SFs) have a major impact on productivity. Productivity is mostly increased by natural fractures (NFs) when the pressure front reaches those. Sun and Wang (2024) numerically investigated the initiation and propagation of hydraulic fractures under in situ load for three multiborehole clusters. A multiscale hydraulic coupling model was constructed by refining the methodology of the conventional pipe domain model within the discrete element approach. The maximum primary stress is the second factor that governs fracture development, after bedding planes, according to the numerical studies. An important factor in controlling fracture connectivity is radial stress, whereas the circumferential stress mostly affects fracture propagation. When multiborehole clusters are hydraulically fractured, the area around the boreholes is vulnerable to stability loss, and the stress shadow has a big impact on how the fracture network is formed.

Bhattacharyya and Vyas (2022d) designed an ML-based reservoir simulator to expedite the traditional reservoir simulation by

reducing the number of iterations at each time step in a single gas-phase reservoir. In this study, the proposed methodology was successfully applied to expedite the 3-phase black oil reservoir simulation, demonstrating that the application of this ML-based methodology is not limited to a single synthetic model.

Objective

In this study, an ML-based 3-phase black oil reservoir simulator was built that is capable of reducing the number of iterations at each time step, thereby reducing the total computational time required for the entire simulation run. This developed simulator uses traditional finite-difference fluid flow equations during simulation.

Reservoir Model Description

The model consists of a 3-phase black oil fully implicit reservoir simulator. The reservoir model consists of a Cartesian grid with $25 \times 25 \times 25$ grid blocks, and each grid block size is $20 \text{ ft} \times 20 \text{ ft} \times 20 \text{ ft}$. The rock permeability and porosity are randomly varied in the grid blocks. In this model, the upper third is filled with gas, the middle third is filled with oil, and the bottom third is filled with water. To maintain pressure in the reservoir, water injection is done in the lower southwest corner, and fluid production is obtained from the middle northeast corner.

Methodology

A brief explanation of the random forest (RF) model is given in the following section.

Random Forest

The random forest (RF) is an ensemble-based ML algorithm that uses a large number of uncorrelated decision trees. A subsample of predictor variables and a bootstrap subsample of training data are collected from the training data using replacement to model each regression tree. The residual sum of squares (RSS) at each node is minimized through repeated division of the data variable space to construct a regression tree. The final prediction is given by the averaged responses (Breiman 2001).

Eqs. (1) and (2) express the RSS as follows:

$$RSS = \sum_{c=1}^n \sum_{i=1}^{n_c} (y_i - m_c)^2 \quad (1)$$

$$m_c = \frac{1}{n_c} \sum_{i=1}^{n_c} y_i \quad (2)$$

The RSS is minimized by making splits at each node. This is achieved by comparing multiple split possibilities using various variables and various split points in those variables. Once a node is split, it is divided into two nodes. This process continues until the number of data points in a node reaches a certain predetermined limit, after which the process stops (Breiman 2001).

The RF, being an ensemble-based technique, employs model averaging to determine the final output. Therefore, the overfitting error is reduced to the greatest extent possible.

Generalized Likelihood Uncertainty Estimation

Model averaging was employed using the generalized likelihood uncertainty estimation (GLUE) ensemble technique with

cross-validation using k -fold validation. A model is evaluated in one of the folds (k -folds), which are subgroups of data points, while the remaining folds are used for learning. The evaluation uses many tuning parameter combinations. As a result, one training data set will yield numerous models. Weights for each of these models are determined using the root-mean-square error (RMSE). The weighted average of the responses from these models is the used to estimate the final output for the test data (Bhattacharyya and Vyas 2022a; Mishra 2012).

Partial Differential Equation and Discretization Form for Simulation

Multiphase flow in petroleum reservoirs refers to the simultaneous flow of at most three phases (oil, water, and gas) that convey numerous fluid components. Under ordinary conditions, black-oil systems contain three fluid components: oil, water, and gas. If there is no mass transfer across phases, each fluid component remains within its own phase. The oil and water components are believed to be immiscible; therefore, no mass transfer occurs between these phases. The gas component is believed to be soluble in oil but not in water; therefore, mass transfer happens between the oil and gas phases. The solution gas refers to the gas component that is dissolved in the oil phase. To derive a general mass-conservation equation, the gas component is divided into two components: free gas (in the gas phase) and solution gas (in the oil phase) (Ertekin et al. 2001).

Eqs. (3)–(5) show the general partial differential equation for 3-phase black oil:

$$\frac{\partial}{\partial x} \left(\beta_c \frac{k_x A_x k_{ro}}{\mu_o B_o} \frac{\partial p_o}{\partial x} \right) \Delta x + \frac{\partial}{\partial y} \left(\beta_c \frac{k_y A_y k_{ro}}{\mu_o B_o} \frac{\partial p_o}{\partial y} \right) \Delta y + \frac{\partial}{\partial z} \left(\beta_c \frac{k_z A_z k_{ro}}{\mu_o B_o} \frac{\partial p_o}{\partial z} \right) \Delta z = \frac{V_b}{\alpha_c} \frac{\partial}{\partial t} \left(\frac{\varphi S_o}{B_o} \right) - q_{osc} \quad (3)$$

$$\frac{\partial}{\partial x} \left(\beta_c \frac{k_x A_x k_{rw}}{\mu_w B_w} \frac{\partial p_w}{\partial x} \right) \Delta x + \frac{\partial}{\partial y} \left(\beta_c \frac{k_y A_y k_{rw}}{\mu_w B_w} \frac{\partial p_w}{\partial y} \right) \Delta y + \frac{\partial}{\partial z} \left(\beta_c \frac{k_z A_z k_{rw}}{\mu_w B_w} \frac{\partial p_w}{\partial z} \right) \Delta z = \frac{V_b}{\alpha_c} \frac{\partial}{\partial t} \left(\frac{\varphi S_w}{B_w} \right) - q_{wsc} \quad (4)$$

$$\frac{\partial}{\partial x} \left(\beta_c \frac{k_x A_x k_{rg}}{\mu_g B_g} \frac{\partial p_g}{\partial x} + \beta_c \frac{k_x A_x k_{ro} R_s}{\mu_o B_o} \frac{\partial p_o}{\partial x} \right) \Delta x + \frac{\partial}{\partial y} \left(\beta_c \frac{k_y A_y k_{rg}}{\mu_g B_g} \frac{\partial p_g}{\partial y} + \beta_c \frac{k_y A_y k_{ro} R_s}{\mu_o B_o} \frac{\partial p_o}{\partial y} \right) \Delta y + \frac{\partial}{\partial z} \left(\beta_c \frac{k_z A_z k_{rg}}{\mu_g B_g} \frac{\partial p_g}{\partial z} + \beta_c \frac{k_z A_z k_{ro} R_s}{\mu_o B_o} \frac{\partial p_o}{\partial z} \right) \Delta z = \frac{V_b}{\alpha_c} \frac{\partial}{\partial t} \left(\frac{\varphi S_g}{B_g} + \frac{\varphi R_s S_o}{B_o} \right) - q_{gsc} \quad (5)$$

Oil/Water/Gas Flow Model

The mathematical formulation for multiphase flow in petroleum reservoirs includes flow equations for all fluid components in the reservoir, as well as any extra relationships required to complete the flow description and beginning and boundary conditions. The flow equations are formed by combining the relevant forms of the mass-conservation equation, the equation of state, and Darcy's law. Additional correlations include the phase-saturation limitation and capillary pressures as a function of phase saturation. To solve the mathematical model, initial and boundary conditions must be met. The black oil model refers to the 3-phase oil, water, and gas flow model. Under ordinary conditions, this model consists of three fluid

components: oil, water, and gas. While oil and water are immiscible, gas can exist as free or the solution gas. The reservoir is supposed to be constant in temperature, with fluids in thermodynamic equilibrium throughout. The wetting phase is water, the nonwetting phase is gas, and the intermediate wetting phase is oil

$$p_w = p_o - P_{cow} \quad (6)$$

$$p_g = p_o + P_{cgo} \quad (7)$$

$$S_o = 1 - S_w - S_g \quad (8)$$

where

$$\begin{aligned} \frac{\partial}{\partial x} \left(\beta_c \frac{k_x A_x k_{ro}}{\mu_o B_o} \frac{\partial p_o}{\partial x} \right) \Delta x + \frac{\partial}{\partial y} \left(\beta_c \frac{k_y A_y k_{ro}}{\mu_o B_o} \frac{\partial p_o}{\partial y} \right) \Delta y \\ + \frac{\partial}{\partial z} \left(\beta_c \frac{k_z A_z k_{ro}}{\mu_o B_o} \frac{\partial p_o}{\partial z} \right) \Delta z = \frac{V_b}{\alpha_c \Delta t} \left(\frac{\varphi(1 - S_w - S_g)}{B_o} \right) - q_{osc} \end{aligned} \quad (9)$$

$$\begin{aligned} \frac{\partial}{\partial x} \left(\beta_c \frac{k_x A_x k_{rw}}{\mu_w B_w} \left(\frac{\partial p_o}{\partial x} - \frac{\partial P_{cow}}{\partial x} \right) \right) \Delta x \\ + \frac{\partial}{\partial y} \left(\beta_c \frac{k_y A_y k_{rw}}{\mu_w B_w} \left(\frac{\partial p_o}{\partial y} - \frac{\partial P_{cow}}{\partial y} \right) \right) \Delta y \\ + \frac{\partial}{\partial z} \left(\beta_c \frac{k_z A_z k_{rw}}{\mu_w B_w} \left(\frac{\partial p_o}{\partial z} - \frac{\partial P_{cow}}{\partial z} \right) \right) \Delta z \\ = \frac{V_b}{\alpha_c \Delta t} \left(\frac{\varphi S_w}{B_w} \right) - q_{wsc} \end{aligned} \quad (10)$$

$$\begin{aligned} \frac{\partial}{\partial x} \left(\beta_c \frac{k_x A_x k_{rg}}{\mu_g B_g} \left(\frac{\partial p_o}{\partial x} + \frac{\partial P_{cgo}}{\partial x} \right) + \beta_c \frac{k_x A_x k_{ro} R_s}{\mu_o B_o} \frac{\partial p_o}{\partial x} \right) \Delta x \\ + \frac{\partial}{\partial y} \left(\beta_c \frac{k_y A_y k_{rg}}{\mu_g B_g} \left(\frac{\partial p_o}{\partial y} + \frac{\partial P_{cgo}}{\partial y} \right) + \beta_c \frac{k_y A_y k_{ro} R_s}{\mu_o B_o} \frac{\partial p_o}{\partial y} \right) \Delta y \\ + \frac{\partial}{\partial z} \left(\beta_c \frac{k_z A_z k_{rg}}{\mu_g B_g} \left(\frac{\partial p_o}{\partial z} + \frac{\partial P_{cgo}}{\partial z} \right) + \beta_c \frac{k_z A_z k_{ro} R_s}{\mu_o B_o} \frac{\partial p_o}{\partial z} \right) \Delta z \\ = \frac{V_b}{\alpha_c \Delta t} \left(\frac{\varphi S_g}{B_g} + \frac{\varphi R_s (1 - S_w - S_g)}{B_o} \right) - q_{gsc} \end{aligned} \quad (11)$$

The three unknowns, p_o , S_w and S_g are evaluated by solving Eqs. (9)–(11) and the solutions are substituted in Eqs. (6)–(8) to calculate remaining three unknowns, p_w , p_g , and S_o .

Discretization

Discretization is a technique of approximating a differential equation using a collection of finite-difference equations. The discretization method involves replacing the derivatives in the differential equation with algebraic approximations. For time-dependent issues, the solution to a differential equation is continuous in both space and time. The solution to finite-difference equations is defined exclusively for a finite collection of discrete points in space and discrete time levels.

The discretization of Eqs. (9)–(11) is performed as follows:

$$\Delta[T_o * \Delta p_o] = \frac{V_b}{\alpha_c \Delta t} \Delta_t \left(\frac{\varphi(1 - S_w - S_g)}{B_o} \right) - q_{osc} \quad (12)$$

$$\Delta[T_w * (\Delta p_o - \Delta P_{cow})] = \frac{V_b}{\alpha_c \Delta t} \Delta_t \left(\frac{\varphi S_w}{B_w} \right) - q_{wsc} \quad (13)$$

$$\begin{aligned} \Delta[T_g * (\Delta p_o + \Delta P_{cgo}) + T_o * R_s * \Delta p_o] \\ = \frac{V_b}{\alpha_c \Delta t} \left(\Delta_t \left(\frac{\varphi S_g}{B_g} \right) + \Delta_t \left[\frac{\varphi R_s (1 - S_w - S_g)}{B_o} \right] \right) - q_{gsc} \end{aligned} \quad (14)$$

Eqs. (15)–(17) show the implicit finite-difference approximations for all the grid blocks

$$\begin{aligned} \Delta[T_o * \Delta p_o]^{n+1} \frac{V_b}{\alpha_c \Delta t} \left\{ \left(\frac{\varphi(1 - S_w - S_g)}{B_o} \right)^{n+1} \right. \\ \left. - \left(\frac{\varphi(1 - S_w - S_g)}{B_o} \right)^n \right\} - q_{osc}^{n+1} \end{aligned} \quad (15)$$

$$\begin{aligned} \Delta[T_w * (\Delta p_o - \Delta P_{cow})]^{n+1} \frac{V_b}{\alpha_c \Delta t} \left\{ \left(\frac{\varphi S_w}{B_w} \right)^{n+1} - \left(\frac{\varphi S_w}{B_w} \right)^n \right\} \\ - q_{wsc}^{n+1} \end{aligned} \quad (16)$$

$$\begin{aligned} \Delta[T_g * (\Delta p_o + \Delta P_{cgo}) + T_o * R_s * \Delta p_o]^{n+1} \\ = \frac{V_b}{\alpha_c \Delta t} \left\{ \left(\frac{\varphi S_g}{B_g} + \frac{\varphi R_s (1 - S_w - S_g)}{B_o} \right)^{n+1} \right. \\ \left. - \left(\frac{\varphi S_g}{B_g} + \frac{\varphi R_s (1 - S_w - S_g)}{B_o} \right)^n \right\} - q_{gsc}^{n+1} \end{aligned} \quad (17)$$

Linearization

The continuous partial differential equations that describe multi-phase flow problems in porous media are severely nonlinear. These equations are discretized into a set of coupled, nonlinear finite-difference equations, which must be linearized. To linearize strongly nonlinear variables caused by saturation-dependent features, various approaches are applied to multiphase-flow situations.

Fully Implicit Method or Newton's Iteration

The partial derivatives with respect to the unknowns are evaluated as tangents at the latest iterations for the upstream points $i \pm 1$ and i with simple iteration. For example, at p_o^{n+1} and at iteration level v , the time derivative coefficients are linearized.

$$\begin{aligned} T_{ox, i \pm 1}^{n+1} (p_{o, i \pm 1}^{n+1} - p_{o, i}^{n+1}) \\ \approx T_{ox, i \pm 1}^{v+1} (p_{o, i \pm 1}^{v+1} - p_{o, i}^{v+1}) \\ \approx T_{ox, i \pm 1}^{v+1} (p_{o, i \pm 1}^v - p_{o, i}^v) + T_{ox, i \pm 1}^v (\delta p_{o, i \pm 1} - \delta p_{o, i}) (p_{o, i \pm 1}^v - p_{o, i}^v) \\ \times \left[\sum_{l=w, g} \left(\frac{\partial T_{ox}}{\partial S_l} \right)_{\pm}^v \delta S_{l, \pm} + \left(\frac{\partial T_{ox}}{\partial p_o} \right)_{\pm}^v \delta p_{o, \pm} \right] \end{aligned} \quad (18)$$

$$\left(\frac{\partial T_{ox}}{\partial S_l} \right)_{\pm}^v = G_{f_{p_o}}^v \frac{\partial f_{so}^v}{\partial S_l} \quad (19)$$

$$\left(\frac{\partial T_{ox}}{\partial p_o} \right)_{\pm}^v = G_{f_{p_o}}^v \frac{\partial f_{p_o}^v}{\partial p_o} \quad (20)$$

$$\delta X = X^{v+1} - X^v \quad (21)$$

Table 1. The computational time reduction achieved through the application of the novel machine learning (ML)-based reservoir

Sl No.	Type of simulator	Total time taken to perform the simulation run (s)	Total amount of computational time saved (s)
1.	Reservoir simulation without ML	1,990	1,270
2.	Reservoir simulation with ML	720 (training time = 60 s)	

$$\begin{aligned} \Delta_x [T_{px} \Delta_x p_o]_i^{n+1} &= T_{px+}^v [p_{o_{i+1}}^v - p_{o_i}^v] + T_{px-}^v [p_{o_{i-1}}^v - p_{o_i}^v] + T_{px+}^v [\delta p_{o_{i+1}}^v - \delta p_{o_i}^v] \\ &+ T_{px-}^v [\delta p_{o_{i-1}}^v - \delta p_{o_i}^v] + \sum_{l=w,g} [p_{o_{i+1}}^v - p_{o_i}^v] \left(\frac{\partial T_{px}}{\partial S_l} \right)_+^v \partial S_{l+} \\ &+ \sum_{l=w,g} [p_{o_{i-1}}^v - p_{o_i}^v] \left(\frac{\partial T_{px}}{\partial S_l} \right)_-^v \partial S_{l-} \\ &+ [p_{o_{i+1}}^v - p_{o_i}^v] \left(\frac{\partial T_{px}}{\partial p_o} \right)_+^v \partial p_{o+} + [p_{o_{i-1}}^v - p_{o_i}^v] \left(\frac{\partial T_{px}}{\partial p_o} \right)_-^v \partial p_{o-} \end{aligned} \quad (22)$$

Newton's Method or Simultaneous Solution Method

The simultaneous solution (SS) approach aims to solve all equations for multiphase flow simultaneously

$$[J]^v \delta \vec{X} = -\vec{R}^v \quad (23)$$

$$v = 0, 1, 2, \dots$$

$$n = 1, 2, 3, \dots, N$$

$$\delta \vec{X} = \vec{X}^{v+1} - \vec{X}^v; \quad \vec{X}^0 - \vec{X}^n \quad (24)$$

$$\vec{R} = (\vec{R}_1, \vec{R}_2, \vec{R}_3, \dots, \vec{R}_N)^T \quad (25)$$

$$\vec{R}_n = (\vec{R}_{w_n}, \vec{R}_{g_n}, \vec{R}_{o_n})^T \quad (26)$$

Eq. (23) can be expressed as

$$\{[F]^v - [C]^v + [Q]^v\} \delta \vec{X} = -\{\vec{F}^v - [\vec{C}^v - \vec{C}^n] \vec{Q}^v\} \quad (27)$$

where F corresponds to the flow terms, C corresponds to the accumulation terms, and Q corresponds to the production (injection) terms

$$\vec{X} = (\vec{X}_1, \vec{X}_2, \vec{X}_3, \dots, \vec{X}_N)^T \quad (28)$$

$$\vec{X}_n = (S_{w_n}, S_{g_n}, p_{o_n})^T \quad (29)$$

$$\vec{F} = (\vec{F}_1, \vec{F}_2, \vec{F}_3, \dots, \vec{F}_N)^T \quad (30)$$

$$\vec{C} = (\vec{C}_1, \vec{C}_2, \vec{C}_3, \dots, \vec{C}_N)^T \quad (31)$$

$$\vec{Q} = (\vec{Q}_1, \vec{Q}_2, \vec{Q}_3, \dots, \vec{Q}_N)^T \quad (32)$$

$$[J]^v = [[F]^v - [C]^v + [Q]^v]$$

$$\vec{R}^v = \{\vec{F}^v - [\vec{C}^v - \vec{C}^n] + \vec{Q}^v\}$$

The computation of the Jacobian matrix determines the residual, which starts the Newtonian iterations. In the first iteration (i.e., $v = 0$), $\delta \vec{X}$ is evaluated from Eq. (23), and $\vec{X}^{(1)}$ is calculated from Eq. (24). Subsequent iterations proceed until convergence is attained. This procedure is employed as it is unconditionally stable and gives satisfactory results even for large time steps.

Peaceman Model

The Peaceman well model is based on the premise that the pressure calculated for a well block is the same as the flowing pressure at an equivalent radius, r_{eq} (Ertekin et al. 2001):

$$r_{eq} = 0.28 \frac{\{[k_x^{\frac{1}{2}} \Delta x^2] + [k_y^{\frac{1}{2}} \Delta y^2]\}}{\frac{k_y^{\frac{1}{2}}}{k_x^{\frac{1}{2}}} + \frac{k_x^{\frac{1}{2}}}{k_y^{\frac{1}{2}}}} \quad (33)$$

Reducing the Number of Iterations

A reservoir simulator performs a large number of iterations while solving equations for the next time level, and these iterations consume a significant portion of the simulation time. In this study, the simulated data gathered at the end of the first 30 days were employed to build an ML model. Before the beginning of the next time step, the ML model was employed to predict the fluid properties, including the formation volume factor, fluid viscosity, fluid density, and relative permeability. The reservoir simulator requires fewer iterations and time while solving for the pressure and saturation of the next time level, as the fluid properties and relative permeability predicted by the ML model for the next time step are very close to the actual values. This results in a significant reduction in the overall computational time involved in the entire simulation run in the developed reservoir simulator and saves a large amount of computational time. Thus, the reservoir simulation process becomes much faster and computationally more efficient, as shown in Table 1.

The following steps are employed for reducing the number of iterations:

1. In the initial stage, reservoir simulation was performed for a period of 30 days without employing any ML modeling. At the end of the 30 days, the data set gathered from Day 1 to Day 30 was employed to build the initial ML model. The time step size

Table 2. Input variables, output variables, and the machine learning (ML) algorithm used for developing the ML model

ML algorithm	Input parameters to the ML model	Output parameters from the ML model
Random forest	Pressure and saturations in the grid blocks (at the beginning of a given time step) and Transmissibility at the interfaces between the central grid block and the 6 neighboring grid blocks (at the beginning of a given time step)	Pressure and saturation values at the end of a given time step to be utilized to calculate initial estimate of fluid properties like formation volume factor, fluid viscosity and fluid density, and relative permeability, respectively

- has been fixed as 1 day. The tolerance is fixed as the difference between the predicted and target output $\leq 1\%$. For well control, the bottom-hole pressure is fixed at 4,500 psi.
- One ML model is built for each grid block by employing the simulated data (15,625 ML models for 15,625 grid blocks). The grid block's pressure and saturation, the pressure and saturation of the neighboring grid blocks, and the transmissibility across the interfaces with the neighboring grid blocks are the input parameters. The pressure and saturation of the given grid block at the next time level (at the end of current time step) are the output parameters for the developed ML model for a given grid block. The pressure values predicted by this ML model were employed for predicting the fluid properties, including the formation volume factor, fluid viscosity and fluid density, and the saturation values predicted by this ML model were employed for predicting the relative permeability. This process was repeated for all the grid blocks in the entire reservoir model.
 - Fluid properties were calculated by employing the pressure and relative permeability, as predicted by the ML model, to solve for the pressure and saturation for the next time level, instead of using the starting values of fluid properties and relative permeability based on the previous time level. The fully implicit equations were solved to determine the pressure and saturation values, which were then updated for each grid block.
 - On completion of the most recent time step of the simulation process, the data gathered during the most recent time step were employed to train and update the developed ML model for each of the grid blocks.

- Repeat the process from Steps 2 to 4 for all the subsequent time steps.

Table 2 shows the input variables, output variables, and the ML algorithm used for developing the ML model. The predicted pressure and saturation values by the ML model are used to estimate the initial values of fluid properties and relative permeability, respectively, in each grid block, which are then utilized for calculating the pressure and saturation at the next time level. Since this initial ML-based estimate is close to the actual values, the number of iterations during each time step is reduced.

The reservoir model is in 3D; therefore, the grid block system consists of a central grid block surrounded by 6 neighboring grid blocks, culminating in a 7-grid block system.

The hyperparameters employed for designing the RF-based model are as follows:

- This model uses a maximum of 100 decision trees.
- The model is evaluated using the RMSE metric.
- In a decision tree, the maximum depth is permitted until purity is attained.
- Each node can have a maximum of 6 splits.

Results and Discussion

Table 1 shows the total computational time required before and after applying the developed ML model.

The simulation execution time reduction factor (SETRF) is calculated as follows:

$$\text{SETRF} = \frac{\text{Computational time taken before the application of the developed ML model}}{\text{Computational time taken after the application of the developed ML model}} = \frac{1,990 \text{ s}}{720 \text{ s}} = 3$$

From Table 1, it is observed that the total computational time involved in the entire simulation run can be reduced by a factor of three. That is, the simulation process can be made three times faster through the application of this novel ML-based reservoir simulator. The results indicate that the application of this developed reservoir simulator can make the traditional reservoir simulation process computationally more efficient while preserving the standard fluid flow physics involved.

Figs. 1–5 show the pressure saturation maps, gas production, oil production, cumulative gas production, and cumulative oil production, respectively. The results predicted by the ML-based reservoir simulator are compared with those predicted by the KAPPA Workstation-Rubis to evaluate the accuracy of the developed reservoir simulator. The KAPPA Workstation-Rubis performs numerical simulation.

In Fig. 1, the green portion represents the oil saturation, the blue portion represents the water saturation, and the red portion represents the gas saturation. Due to the water injection near the bottom of the southwest corner, the grid blocks near it are more pressurized than the grid blocks situated over the rest of the reservoir model, as shown in Fig. 1.

The following conclusions can be drawn after observing the Figs. 2–5:

Fig. 2 shows the comparison between the developed ML-based reservoir simulator and KAPPA workstation-Rubis results for cumulative gas production. The figure shows that the results are closely matching. From the figure, the producer began producing oil, causing the reservoir pressure inside the reservoir to decline, and the

compressed gas to expand. Since the mobility of gas is significantly greater than that of oil, an abrupt increase in gas production accompanied by minimal oil production was observed. When water injection commenced after 175 days and the reservoir pressure was maintained, the remaining gas was depleted at a nearly constant rate, accompanied by an increase in oil production. Fig. 3 compares the results of the developed ML-based reservoir simulator and KAPPA Workstation-Rubis for cumulative oil production. It can be observed that the results are closely matching. Initially, due to reservoir pressure depletion and a rapid increase in gas production, a lower amount of oil was produced. When reservoir pressure maintenance began through water injection after 175 days, there was a rapid increase in oil production, accompanied by a constant rate of gas depletion. Fig. 4 shows a comparison between the developed ML-based reservoir simulator and KAPPA Workstation-Rubis results for gas production rates, indicating that the results closely match. The gas production rate shows an increasing trend, which is further verified by the KAPPA Workstation-Rubis results. Fig. 5 compares the developed ML-based reservoir simulator and KAPPA Workstation-Rubis results for the oil production rate. Once again, the results closely match. The oil production rate shows an increasing trend, which is further verified by the KAPPA Workstation-Rubis results.

Limitations

The primary aim of this study was to prove the applicability of the proposed method. Therefore, a synthetic 3-phase black oil scale

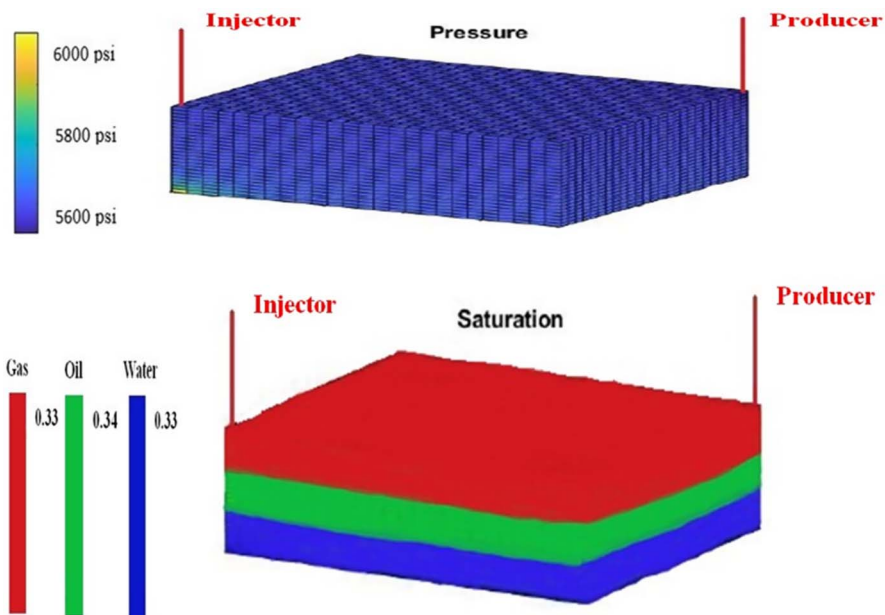


Fig. 1. Pressure and saturation maps of the reservoir.

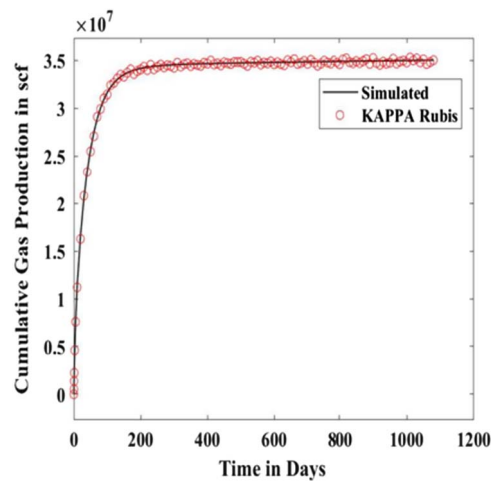


Fig. 2. Cumulative gas production.

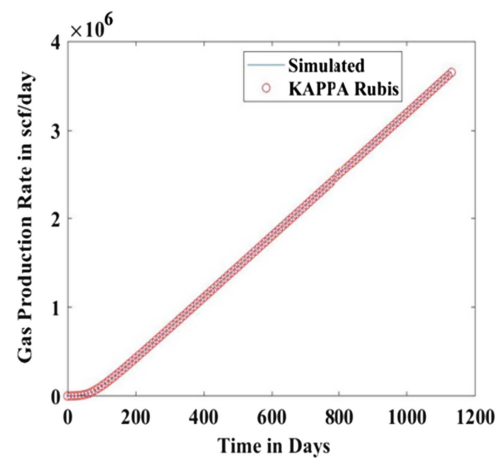


Fig. 4. Gas production rate.

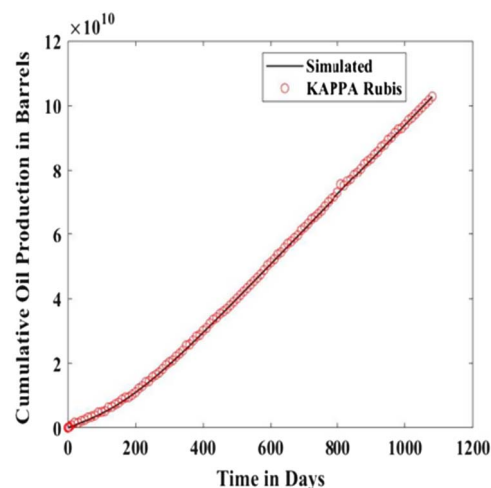


Fig. 3. Cumulative oil production.

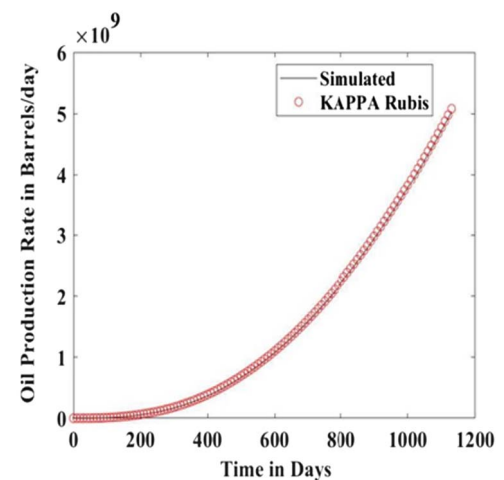


Fig. 5. Oil production rate.

model was chosen to prove the applicability of the proposed method. This study can be extended, and the applicability of the developed ML-based methodology can be tested on real field models such as Society of Petroleum Engineers (SPE) standard models with complex geological and petrophysical characteristics.

Conclusions

The developed ML-based methodology was successfully applied to increase the speed of traditional reservoir simulation by a magnitude of 3 while conserving the standard fluid flow physics involved. Random Forest based ML model has been tested in this study to successfully reduce number of iterations during conventional reservoir simulation process thereby reducing computational time involved during reservoir simulation. The accuracy of the developed simulator has also been matched well with commercial simulator KAPPA-Rubis for the synthetic model used. This proposed methodology particularly becomes useful when a fast and accurate prediction of the reservoir production behavior is required within the limited time period and limited computational resources with a large number of grid blocks involved. The applicability of the ML-based method can be tested on real field models such as SPE standard models with complex geological and petrophysical characteristics.

Data Availability Statement

All data, models, and code generated or used during the study appear in the published article.

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Author Contributions

Subhrajyoti Bhattacharyya: Conceptualization; Data curation; Formal analysis; Funding acquisition; Investigation; Methodology. Aditya Vyas: Conceptualization; Data curation; Formal analysis; Funding acquisition; Investigation; Methodology.

Notation

The following symbols are used in this paper:

- A_x = cross-sectional area in x -direction, ft^2 ;
- A_y = cross-sectional area in y -direction, ft^2 ;
- A_z = cross-sectional area in z -direction, ft^2 ;
- B_g = gas formation volume factor, rb/scf ;
- B_o = oil formation volume factor, rb/stb ;
- B_w = water formation volume factor, rb/stb ;
- c = number of f nodes;
- i = superscript for i th gridblock;
- $i \pm 1$ = superscript for $i \pm 1$ th gridblock;
- $[J]$ = Jacobian matrix;
- k_{rg} = relative rock permeability to gas, dimensionless;
- k_{ro} = relative permeability to oil, dimensionless;
- k_{rw} = relative permeability to water, dimensionless;

- k_x = permeability in x -direction, millidarcy;
- k_y = permeability in y -direction, millidarcy;
- k_z = permeability in z -direction, millidarcy;
- m_c = mean of the observed values, y_i ;
- n = superscript for current time level;
- $n + 1$ = superscript for next time level;
- n_c = number of data points in a node;
- P_{cgo} = capillary pressure of gas with respect to oil, psi;
- P_{cow} = capillary pressure of oil with respect to water, psi;
- p_g = pressure of gas, psi;
- p_o = pressure of oil, psi;
- p_w = pressure of water, psi;
- q_{gsc} = gas flow rate at standard conditions, scf/day ;
- q_{osc} = oil flow rate at standard conditions, stb/day ;
- q_{wsc} = water flow rate at standard conditions, stb/day ;
- R = residual of the original equation;
- R_s = solution gas-oil ratio, scf/stb ;
- S_g = saturation of gas, fraction;
- S_o = saturation of oil, fraction;
- S_w = saturation of water, fraction;
- T_g = transmissibility of gas at the interfaces of the grid block, scf/day-psi ;
- T_o = transmissibility of oil at the interfaces of the grid blocks, stb/day-psi ;
- T_w = transmissibility of water at the interfaces of the grid blocks, stb/day-psi ;
- V_b = bulk reservoir volume, ft^3 ;
- y_i = observed or actual response value;
- Δ = partial-difference operator;
- δ = delta operator;
- β_c = transmissibility conversion factor = 1.127×10^{-3} ;
- φ = porosity, %;
- μ_g = viscosity of gas, centipoise;
- μ_o = viscosity of oil, centipoise;
- μ_w = viscosity of water, centipoise;
- v = superscript for current iteration;
- $v + 1$ = superscript for next iteration; and
- $\delta \vec{X}$ = vector of change in pressure and saturation from one iteration to the next.

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